1. Introduction

Chronic diseases, defined broadly as conditions that last one year or longer and require ongoing medical attention and/or limit activities in daily living, is a major public health concern in the United States [1]. In 2018, the National Health Interview Survey estimated that 129 million Americans have at least one major chronic disease [2]. This study examines demographic and geographic factors that predict chronic disease risk factors. By implementing our models, we aim to extract meaningful insights that can better inform patients and the general public about chronic disease risk.

2. Problem Definition

We want to predict an individual's risk of developing or dying from one of five major chronic diseases: diabetes, cardiovascular disease, chronic kidney disease (CKD), chronic obstructive pulmonary disease (COPD), and cancer. We will use the U.S. Chronic Disease Indicators dataset from the Centers for Disease Control and Prevention (CDC) merged with state Medicaid expenditures and the U.S. Census Bureau's median household income dataset. After identifying optimal models per disease for predicting an individual's chronic disease risk factor, we want to create an interactive dashboard that displays the risk score for the chronic diseases above filtered by gender, race/ethnicity, median income, Medicaid expenses at the state level, and U.S. geographical location.

3. Literature Survey

Diabetes is characterized by elevated blood glucose levels, which damage the heart, blood vessels, eyes, kidneys, and nerves [3]. Several machine learning models, with K-nearest-neighbors having the highest accuracy of 79.6%, predicted diabetes with several medical variables [4]. Notably, research demonstrated the effectiveness of the Synthetic Minority Over-sampling Technique (SMOTE), used to address class imbalance dataset issues by generating new synthetic data points belonging to the minority class [5]. Noting what methods improve accuracy, a significant limitation for both studies above was using the Pima Indian dataset (768 patients), raising generalizability questions [6]. Another study projected the future prevalence of Type 1 and Type 2 diabetes in youth aged <20 years using the wider SEARCH dataset [7]. This aligns more with our study due to its larger data span and focus on prediction.

Chronic obstructive pulmonary disease (COPD) is an ongoing lung condition caused by damage to the lungs. The damage results in inflammation inside the airways that limit airflow into and out of the lungs. This limited airflow is obstruction [8]. COPD is underdiagnosed, with many patients presenting to tertiary care having already developed advanced-stage disease with a low prognosis, with 30% of those hospitalized dying within 2 years [9]. Prediction models are sparse. A survey of four prediction models for COPD risk assessment using stepwise multiple regression, Cox regression, and Bayesian networks stated that smoking is the only consistent risk factor [10]. Adding more factors alongside smoking can bolster our model's strength. Machine learning algorithms have been utilized for aspects of COPD other than diagnosis. A Cox regression model on clinical and quantitative CT features outperformed models predicting COPD mortality, and COPD exacerbations were indicated by different data types, including triage data, functional respiratory imaging, and wearable device data [11] [12] [13] [14]. These models do not focus on prediction or demographic and socioeconomic factors, unlike ours.

Chronic kidney disease (CKD) is characterized by progressive damage and loss of function in the kidneys [15]. A nested case-control study developed a prediction model identifying individuals most likely to develop CKD with regression and survival analysis results [16]. The study found that age, diabetes, and standard high values of urea nitrogen, TGF-β, and ADMA were independent risk factors. These are medical factors, but like this study, we will compare approaches and select one with high accuracy. Another machine learning model predicted the need for renal replacement therapy and disease progression for patients with stage 3-5 CKD with risk analysis and time-to-event analysis [17]. Though the time-to-event model performed well, it overrepresented the middle-income socioeconomic stratum. Our project uses data from a wide range of socioeconomic statuses. A separate study examined 12 different

classifiers, and the XgBoost classifier was identified as the best-performing model. A 10-fold cross-validation then checked for over-fitting in the findings [18]. The team may implement cross-validation to reduce overfitting.

Cardiovascular disease (CVD) affects the heart and blood vessels [19]. Researchers used deep learning (DL) and ensemble methods to predict CVD risk from electronic health records (EHRs). Comparing performance with traditional risk scores found that biomarkers, genetic information, and social determinants of health (SDOH) enhanced prediction accuracy [20]. Although these methods are useful, this study had inconsistent ICD definitions. We should standardize definitions to ensure consistent coding across merged datasets. The Multi-Ethnic Study of Atherosclerosis (MESA) study used random forests (RF) to predict advanced CVD with biomarkers, imaging data, and clinical measures. Deep phenotyping using these markers led to better outcome prediction [21]. Random forests are powerful but can overfit, especially when using many predictors. We will experiment with feature selection techniques to generalize our model. This study also only focused on middle-aged adults without existing CVD. Another review analyzed various approaches for CVD prediction, including SVM, random forest, KNN, decision trees, and logistic regression, evaluating their analysis of patient demographics, medical history, and diagnostic data [22]. Many studies used smaller, less diverse datasets. We will apply our model to other populations, including those with existing conditions or other age groups.

Cancer is characterized by cells uncontrollably growing and potentially spreading to other body parts [23]. One study predicted future cancer incidence based on age, race, ethnicity, and sex, using a projection equation for twenty-three types to indicate that cancer would increase by 29.5% between 2020 and 2040. [24]. Risk prediction models provide data-driven predictions, but well-known ones heavily lean toward breast cancer. A recent survey compared supervised machine learning techniques: support vector machine (SVM), K-nearest neighbors, random forests, artificial neural networks (ANNs) and logistic regression with the Wisconsin Breast Cancer dataset [25]. Random forests presented higher performance. Models in another study included 5-18 medical risk factors, with age being the only factor in all models. Although new improvements included adding genetic information (SNPs), breast density, and biomarkers, both studies did not incorporate population characteristics and social health determinants [26].

4. Proposed Method:

Although several prediction models have been developed in the last decade to predict the risk of developing chronic diseases, most models focus on a single disease rather than providing a comprehensive health risk assessment. Existing studies also primarily use small datasets, sometimes with a significant predominance of an ethnic group, and often rely heavily on individual clinical markers while overlooking broader socioeconomic and geographic factors. Our approach integrates a CDC state-level chronic disease indicators dataset with 668,502 data points total from 2008 to 2021 with primary factors of race, gender, median income, and Medicaid expenditures. By incorporating both Medicaid expenditure data alongside demographic and geographical factors, our innovative model captures a critical financial dimension that most existing prediction frameworks overlook. Additionally, the dataset's extended time frame of 13 years (2008-2021) provides a more robust foundation for identifying long-term trends that shorter-term studies often miss. The filtering process began by selecting records related to our specific chronic disease. We focused on data where the StratificationCategory1 was either "Gender" or "Race/Ethnicity" and columns such as year, location, topic, question, data value, and stratification information. We removed rows with missing values in the "Number" or "Prevalence" field and converted the year and count/prevalence to integer values for consistency. For the Medicaid expenditures data, we selected relevant years (2011-2020), reshaped it to a long format, and grouped by state and year to calculate total expenses. Finally, we incorporated median household income data from the Census Bureau after cleaning and reshaping it, and joined all three datasets by state and year, removing rows with missing values in the final dataset. **We experimented with** a wide range of machine learning models to find the optimal approach for chronic disease prediction.

For cardiovascular disease prediction, we tried linear regression, random forest, gradient boosting, XGBoost, decision trees, support vector machines, and K-nearest neighbors. These models were assessed based on their ability to capture complex relationships between demographic, socioeconomic, and health-related factors. Additionally, regularization techniques such as ridge and lasso regression were applied to mitigate overfitting and enhance model generalization. **For COPD prediction,** we explored multiple regression techniques, including linear regression, ridge regression, elastic net, and tree-based methods such as random forests. Linear regression provided a strong baseline, capturing general trends between demographic and socioeconomic factors and disease prevalence. Ridge regression improved upon this by reducing overfitting through L2 regularization, ensuring stability in the presence of multicollinearity. Among non-linear models, random forests performed well due to their ability to capture complex interactions. While linear and ridge regression suggest that COPD exhibits a largely linear relationship with socioeconomic factors, incorporating non-linearity through ensemble methods enhances predictive accuracy. **For cancer prediction,** we trained two random forest models: a regression model to predict the number of diagnosed cases and a classification model to predict cancer type. Random forest was selected because its ensemble approach of building multiple decision trees helps mitigate overfitting and they excel at capturing non-linear relationships and complex interactions between variables. **For diabetes prediction,** we used linear modeling, random forest, generalized boosted modeling, and XGBoost. These models were selected for their complementary strengths: linear modeling established baseline relationships between demographic factors and diabetes prevalence; random forest captured nonlinear interactions between variables; generalized boosted modeling offered strong predictive performance through iterative improvement; and XGBoost provided optimization advantages through regularization and parallel processing. **For CKD prediction,** we explored multiple linear regression, random forest, and gradient boosting. Multiple linear regression served as the baseline model and allowed us to explore interpretable coefficients. Gradient boosting let us make predictions by reducing variance and bias. Random forest was selected as the top performing model for making CKD mortality count predictions since our predictive features exhibited non-linear, complex relationships. Random forest accounted for overfitting and handled categorical variables better than linear regression and gradient boosting.

Once an optimal model had been identified for each chronic disease, these were utilized in predicting mortality for years 2021-2050. The predictions for each chronic disease were exported then combined into a master Excel spreadsheet. The spreadsheet was then uploaded into Tableau and a map of the U.S. was generated using the longitude and latitude of the state listed. A Risk Score was calculated by taking the number of deaths and dividing that by 1,000. Users can see the Risk Score for a specific chronic disease based on median income, gender, ethnicity, and year. All of these features are filterable via the interactive dashboard.

5. Evaluation

We aim to determine which machine learning algorithms provide optimal performance when predicting risk factors for diabetes, COPD, CKD, cardiovascular disease, and cancer. We also evaluate which performance metrics are most appropriate for assessing our models. We finally explore insights users can extract from our interactive dashboard.

The cancer experimental testbed consisted of an R environment using tidyverse for data manipulation and visualization with machine learning packages including randomForest, XGBoost, and caret. We preprocessed by converting categorical variables to factors, ensuring proper numeric data types, and removing rows with missing values, then implemented temporal splitting where earlier years (2011-2018) were used for training while later years (2019-2020) were reserved for testing. Two random forest models were developed: a regression model to predict diagnosed cases and a classification model to predict cancer types, both using state, gender, race, Medicaid expenses, and median income as predictors. We

observed that cervical cancer, prostate cancer, and breast cancer had the highest number of cases. We also found that women had more cancer diagnoses than men overall, although more analysis needs to be completed to accurately understand gender-specific, race-specific, and socioeconomic trends. **CKD predictions** used R packages such as dplyr, tidyr, randomForest, and gbm to clean and combine datasets and make predictions. After training and testing models, future predictions fed into an interactive Tableau dashboard, which allowed users to explore how factors such as location, income, and demographics affect CKD risk through 2050. Most deaths fell between 0 and 1,000. Some predictions forecasted deaths above 6,000-8,000, but these were relatively rare. The mean predicted death count by race saw White, Non-Hispanic in the lead followed by Black, Non-Hispanic, American Indian/Alaska Native, Hispanic, and then Asian/Pacific Islander. The mean predicted death count by gender was about 50/50 and median income appears to be inversely related to death counts. Medicaid expenses appeared to have a positive correlation with death counts. California was ranked highest in Medicaid expenses and also reported a high death count. Texas, on the other hand, displayed a wide range of death counts with lower Medicaid expenses. States like Alaska, Vermont, and Delaware have both low death counts and low Medicaid expenses. For the CKD prediction system, there does not appear to be a clear linear trend between Medicaid spending and death count. **COPD predictions** were conducted in a modular fashion through Python, leveraging scikit-learn for implementing linear models and ensemble methods. The dataset was cleaned to focus solely on mortality rates within U.S. States, then split into training and testing sets based on year to avoid data leakage that would cause model overfitting. This dataset incorporated demographic, socioeconomic, and health-related features, with model performance evaluated using metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared. Observations from the COPD predictions showed that lower median income was generally associated with higher predicted death counts, suggesting that socioeconomic factors, including access to healthcare and lifestyle choices, may contribute to COPD risk. While some states measured higher healthcare spending along with higher COPD mortality, it is worth noting that our experimentations did not account for total population, which can skew results and generalizations; model predictions should therefore be treated with caution as our study did not include total population or per capita rates. **Diabetes prediction** was conducted using R with Linear Modeling, Random Forest, Generalized Boosted Modeling, and XGBoost. The dataset was split into a training set (2011–2017) and a test set (2018–2020). The models incorporated Demographics, Median Income, Medicaid Spend, and State as features, with predictive performance assessed using root mean squared error (RMSE), R-squared, and mean absolute error (MAE). Higher Medicaid spending was positively correlated with the number of people diagnosed with Diabetes, while Median Income did not appear to be a good indicator of Diabetes diagnosis as the correlation between Median Income and number of people diagnosed was low. White, non-Hispanic individuals exhibited the highest mean diagnosis, and males had an 18% higher mean diagnosis than females. **Cardiovascular disease prediction** was conducted with experiments in Python with scikit-learn and XGBoost.

The dataset was split into a training set (2011–2018) and a test set (2019–2020) to prevent overfitting. The models incorporated demographic, socioeconomic, and health-related features, with predictive performance assessed using metrics like mean squared error (MSE), R-squared, and mean absolute error (MAE). The findings were incorporated into an interactive dashboard. The histogram of Medicaid expenses showed a highly skewed distribution, with most states having lower expenditures and only a few states with very high spending. Based on exploratory data analysis, states with higher Medicaid spending tended to have higher reported cases of cardiovascular disease, though this relationship was not strictly linear. White, Non-Hispanic individuals exhibited the highest mean predicted diagnosis rates, followed by Black, Non-Hispanic and Hispanic populations. Additionally, lower median income levels correlated with increased cardiovascular disease risk, suggesting that socioeconomic status plays a significant role in disease prevalence. **Ultimately, we hope** that earlier disease detection leads to better health outcomes for all of us. Our major risk is developing an analysis from such a large dataset. Data quality and integrity is a major concern; there are missing and incomplete data points, inconsistent data formats, and historical data

inconsistencies across different healthcare systems. Methodological risks also include selection bias in the source population and spurious correlations due to increased sample sizes. As mentioned before, our payoffs are earlier disease detection and a better understanding of disease patterns and risk factors.

Below is a table summarizing each chronic disease, their optimal predictive model, and the model's RMSE:

Optimal Models RMSE

	Cancer	Diabetes	Cardiovascular	COPD	CKD
Best Model	XGBoost	XGBoost	Random Forest	Random Forest	Random Forest
RMSE	3308.7	1034.6	3314.0832	782.82	1005.6

While MAE and R^2 offer intuitive interpretations—representing average absolute error and proportion of variance explained, respectively—they do not sufficiently penalize large deviations. RMSE, on the other hand, squares the residuals before averaging, thus emphasizing larger errors, which is critical in regression contexts where such deviations have significant consequences. This is especially important in real-world applications such as predicting chronic diseases, where large prediction errors could lead to misdiagnosis or delayed interventions. Furthermore, RMSE retains the same units as the response variable, facilitating direct interpretation of prediction error magnitudes. Consequently, RMSE provides a more sensitive and informative assessment of model accuracy, particularly in applications where minimizing the impact of large residuals is a priority.

6. Conclusions and Discussion

This project successfully identified optimal machine learning models for predicting the number of deaths for cancer, cardiovascular disease, diabetes, chronic kidney disease, and chronic obstructive pulmonary disorder. The prediction results were also loaded successfully into Tableau to forecast a Risk Score for each chronic disease based on filterable features such as geographic and economic factors and Medicaid expenditures by state.

The project was also successful since it provided insight on several things: higher Medicaid spending correlated with higher chronic disease mortality, although the relationship is not always linear; lower median household income was associated with higher risk for COPD and CVD; there were disparities across ethnic groups for CKD and cancer; and the cancer diagnoses rates were highest for cervical, breast, and prostate cancer, with women overall exhibiting higher diagnosis rates.

From the interactive dashboard, users such as hospitals, healthcare centers, and medical providers can develop targeted screening programs and/or preventive programs to combat chronic diseases in states with higher Risk Scores. Additionally, public health officials and state health departments can determine whether a higher spend amount for Medicaid increases or decreases chronic diseases in their state. This would help guide informed decisions on whether or not to allocate more or less funding towards Medicaid in their state.

All team members have contributed a similar amount of effort.

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